The Islamia University of Bahawalpur
Department of Information Technology

INFT-4041 · BSIT · 5th Semester · Fall 2026

**E-Commerce Application Development**

Chapter

# 23

# The Semester Project

*Part VI — Reach, Law and Practice*

| | |
|---|---|
| **Taught in** | Week 16 |
| **Laboratory** | Project presentations |
| **Assessment** | **Project due; presentations** |

**This is a class extract.** It is *Chapter 23* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

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Chapter 23

# The Semester Project



**Learning Outcomes**

By the end of this chapter you will be able to:

- **Assemble** the shop you have built since Chapter 5 into a deliverable with a stated scope.
- **Deploy** it from nothing, on a machine that is not yours, in one command.
- **Measure** your own shop and **report** the figures honestly, including the ones that did not improve.
- **Present** and **defend** the decisions you made, naming what you would do differently.
- **Judge** another team's shop against the same criteria.

**Before You Start**

Everything. Specifically: a shop that stands up (Chapter 5), takes an order (Chapter 10), charges correctly (Chapter 11), is hardened (Chapter 16), carries at least one module you wrote (Chapter 19), and can be restored from backup (Chapter 20).

**Key Terms in This Chapter**

scope • deliverable • reproducible build • baseline • evidence • defence • peer review • handover • known limitation

## 23.1 What Is Being Assessed

HEC's own teaching methodology for this course names *Project* and *Presentations*, and this chapter discharges both. The project is not a new piece of work: **it is the shop you have been adding to since week three**, finished, deployed, measured and defended.

**The one sentence the whole project is marked against**

**Can somebody else bring your shop up from nothing, place an order in it, and check your numbers?**

Everything below follows from that. A shop that only runs on your laptop is not deployed. A claim you cannot reproduce is not a result. A decision you cannot explain is not a decision.

## 23.2 The Deliverable

**Five things, and nothing else**

1. **A repository that builds.** `docker compose up -d --build` on a clean machine produces a working shop with your catalogue in it. If a human step is needed, it is one line in the README and it is tested.
2. **The shop itself**: a real catalogue of at least thirty products with options, categories, prices, weights and images; working shipping and tax for a stated place; a completable checkout; at least one payment method; and the Chapter 16 hardening applied.

3. **At least one module you wrote** (Chapter 19), with its admin half, its language file, and an install step — and a sentence saying what the shop could not do without it.
4. **A measurement report**, two to four pages: what you measured, with what, on what machine, what you found, and what you changed as a result. Figures you did not improve are part of the report.
5. **A ten-minute presentation and a defence.**

**Not wanted:** a redesigned theme as the main contribution, a slide deck longer than the shop, or a feature list. The marks are in whether it works, whether it is measured, and whether you can defend it.

| Assessed on | Weight | What earns it |
|---|---|---|
| It builds and runs elsewhere | 20% | Somebody else runs one command |
| The shop works end to end | 20% | An order can be placed and fulfilled |
| Your own extension | 15% | It installs, removes cleanly, and is needed |
| Hardening and correctness | 15% | The Chapter 16 battery, and the go-live list from Chapter 20 |
| Measurement report | 20% | Reproducible figures, honestly reported |
| Presentation and defence | 10% | You can answer "why" |

## 23.3 The Measurement Report

This is where the course differs from a web-development project, and it is worth being explicit about what a good one looks like.

**Four measurements, chosen because you have already made them once**

1. **A catalogue measurement** (Chapter 6): a page's time and query count against catalogue size, with and without an index you chose. *Report the case where the index did not help, if there is one.*
2. **A page weight measurement** (Chapter 8): bytes and requests, before and after one change, with the change named.
3. **A security measurement** (Chapter 16): the battery before and after your hardening, with the findings you could not close listed and costed.
4. **One of your own choosing**, with a stated question it answers.

**Every figure must name the file that produces it**, exactly as this handout does. A number with no command behind it is an assertion.

**What a good report does that a weak one does not**

**It states the machine.** Appendix A of this handout exists because every timing in it is a fact about one laptop. Yours is a fact about yours.

**It reports the measurement that disagreed with you.** This handout printed three wrong versions of Chapter 6's experiment, the store identifier that made 50,000 products invisible, and three security controls that were applied correctly and did nothing. Every one of those is worth more than a result that went as expected, because it shows the method working.

**It separates what it measured from what it believes.** "The index made the page slower at 50,000 products on this machine" is a measurement. "Our shop is fast" is not.

**It says what it could not measure.** Chapter 8's harness could not see fonts requested by CSS, and said so. Knowing the limits of your instrument is the difference between a report and a brochure.

## 23.4 The Defence

Ten minutes of presentation and then questions. The questions are not a test of memory.

**</> What you will be asked, and what a good answer sounds like**

- **"Why this platform?"** Not "it was in the course". Chapter 4 is the argument; make it yours, with the alternative you rejected and why.
- **"Show me an order going through."** Live, on the running shop. Know what status it lands in and what moved in the database when it did (Chapter 10, Chapter 18).
- **"What does this number mean?"** Pointing at any figure in your report. Be able to say what it counts, what it excludes, and what would change it.
- **"What is still wrong with it?"** The best question and the one most teams are unprepared for. A team that names three real limitations is further along than one that claims none.
- **"What would you do with another month?"** Ordered by measured benefit, not by what would be interesting to build.
- **"Who did what?"** Answered the same way by every member.

**Reviewing another team's shop**

Each team reviews one other, using the same instruments, and submits one page.

1. **Clone and build it.** Time how long until you see a storefront. That number is the review's first finding.
2. **Place an order.** Note every step that confused you.
3. **Run the security battery** against *their* container on *your* machine. This is within scope precisely because you have built it yourself and it is running on your own computer — Chapter 16's rule is about systems you do not control, and a classmate's repository running on your laptop is one you do.
4. **Check one of their numbers.** Run the file they name, and report what you got.
5. **Write one page**: what works, what you could not get working, one thing you will steal, and one thing you would change.

**Reproducing somebody else's number is the single most valuable hour of the semester**, in both directions. It is also the only part of this course that cannot be done alone.

## 23.5 A Realistic Schedule

| Week | Milestone | What is actually due |
|---|---|---|
| 10 | Proposal | One page: what you sell, to whom, where you deliver, which payment methods, and the one feature you will build |
| 12 | Catalogue complete | 30+ real products with weights and images, and your first measurement |
| 14 | Your module works | Installs and removes cleanly on a clean container |
| 15 | Hardened and deployed | The go-live list from Chapter 20, and a restore you have timed |
| 16 | Present and defend | And review another team |

**Common Pitfalls**

- **Leaving deployment to the last week.** It is the deliverable with the most ways to fail and the least room to recover.
- **A shop that runs only on the machine it was built on.**
- **Measuring once, at the end.** A baseline you did not take in week 10 cannot be taken in week 16.
- **Reporting only the figures that improved.**
- **Quoting a number without the command that produces it.**
- **Spending the project on the theme.** It is the most visible work and the least of the marks.
- **Twelve demo products with no weights**, which breaks shipping in a way Chapter 11 told you about.
- **A module that cannot be removed**, which means it was never tested from a clean state.
- **Arriving at the defence unable to say what is still wrong with it.**
- **Running anybody's security battery against anything but your own or your classmate's container on your own machine** (Chapter 16).

**Try It Yourself: Ship it**

```
# --- 1. Prove it builds from nothing --------------------------------
cd code
docker compose down -v
docker compose up -d --build
python waitfor.py
# Time this. It is the first number in your report.

# --- 2. Place an order on the fresh shop ----------------------------
python m10_checkout.py --repeat 1
python m10_checkout.py --states
# If this fails on a clean build, your README is wrong.

# --- 3. Install your own extension ----------------------------------
python m19_modules.py --install
python m19_modules.py --measure

# --- 4. Harden, and record the battery before and after -------------
python m16_security.py
```

```
# --- 5. Take the four measurements ---------------------------------
python m06_catalogue.py --quick
python m08_pageweight.py --compare
python m20_deploy.py --load
# Plus your own. Save every output; it is your evidence.

# --- 6. Back it up, destroy it, restore it -------------------------
python m20_deploy.py --backup
# Then do it for real: down -v, up, restore, and check the catalogue
# is there. Record the restore time.

# --- 7. Write the README somebody else will follow -----------------
# Hand it to a classmate and watch them. Every question they ask is
# a line missing from it. Do not help them; write it down.

# --- 8. Write the report -------------------------------------------
# Four measurements, the machine, what changed, what did not, and
# what you could not measure.

# --- 9. Prepare three limitations ----------------------------------
# Real ones, with what each would cost to fix. You will be asked.
```

**What to notice.** Step 7 is the one that decides the first 20% of the marks, and watching somebody fail to follow your README is the only way to write one. Every course does this badly; this is your chance not to.

### Checkpoint — Test Yourself

1. Your shop runs perfectly on your laptop and a classmate cannot get it started. List the five most likely causes in order, say how you would find out which it is without access to their machine, and name the one thing you should have done in week 12 that would have prevented all five.
2. You measured your category page at 180 ms before an index and 195 ms after. Say what you would put in the report, what you would investigate first, and why this result is worth more marks than one that went the way you expected.
3. At the defence you are asked "what is still wrong with it?". Give three answers for the shop as this handout left it, each with the chapter that found it and the cost of fixing it.

### Chapter Summary

- The project is the shop you have built since Chapter 5, finished and defended — not a new piece of work.
- One sentence decides it: can somebody else bring it up from nothing, place an order, and check your numbers?
- Five deliverables: a repository that builds, a shop that works end to end, an extension you wrote, a measurement report, and a defence.
- Every figure names the file that produces it, every timing names the machine, and the measurements that disagreed with you are worth more than the ones that did not.
- A good report separates what it measured from what it believes, and says what its instrument could not see.
- The hardest defence question is "what is still wrong with it?". A team naming three real limitations is ahead of one claiming none.
- Reproducing another team's number is the most valuable hour of the semester, and the only part of the course that cannot be done alone.

- Deploy early. A baseline not taken in week 10 cannot be taken in week 16.

## Review Questions

1. **(MCQ)** The largest single share of the project mark is for: (a) the theme; (b) the number of features; (c) that it builds and runs on another machine; (d) the length of the presentation.
2. **(MCQ)** A measurement in your report that did not improve should be: (a) omitted; (b) re-run until it does; (c) reported with what you think caused it; (d) replaced with an estimate.
3. **(Short)** Explain why a baseline must be taken early, with an example from this course.
4. **(Short)** Give two things a measurement report must state about the machine, and say why each matters.
5. **(Short)** Why is being able to name your shop's limitations worth marks, rather than costing them?
6. **(Applied)** Write the week-10 proposal for a shop selling a product of your choosing in Pakistan: what you sell, to whom, where you deliver, which payment methods and why, the tax configuration, and the one feature you will build with the reason the shop needs it.
7. **(Applied)** You have one month after the semester to take the shop to real customers. Give your ordered plan, each step justified by a measurement or a chapter from this course, and state the three things you would refuse to launch without.